R1.3 Energy from fuels

Incomplete combustion of organic compounds

Learning outcomes

By the end of this section you should be able to:

- Deduce equations for the incomplete combustion of hydrocarbons and alcohols.
- Compare and contrast complete combustion with incomplete combustion of hydrocarbons and alcohols.

Many industries, places of work, and homes make use of the combustion of hydrocarbons. This can be to produce energy, for cooking, or in colder climates, as a source of heat. For safety reasons, buildings must be equipped with carbon monoxide detectors. How is it that the combustion of hydrocarbons can lead to the production of carbon monoxide, and why do we need to be concerned about such by-products?

When organic compounds undergo combustion in low-oxygen environments, they produce not only water and carbon dioxide gas, but other by-products. Depending on the amount of available oxygen, carbon monoxide gas and even carbon particles will be formed. This type of reaction is known as incomplete combustion. The lack of oxygen is often caused by the combustion reaction occurring too quickly, depleting the quantity of available oxygen around the burning fuel. Other factors that lead to incomplete combustion include, limited oxygen concentrations, low combustion temperatures, large fuel particles, and even the presence of water in solid fuels (like wood).

If the amount of oxygen gas is somewhat limited, carbon monoxide gas will form with water. Whereas if the amount of oxygen gas is extremely limited solid particles of carbon will form with water.

When observing a combustion reaction, we can often tell that the reaction is undergoing incomplete combustion due to it releasing smoke, and producing an orange or yellow flame. When a reaction has sufficient oxygen, the flame produced is much higher in temperature and appears blue in colour. As well, complete combustion reactions do not produce smoke.

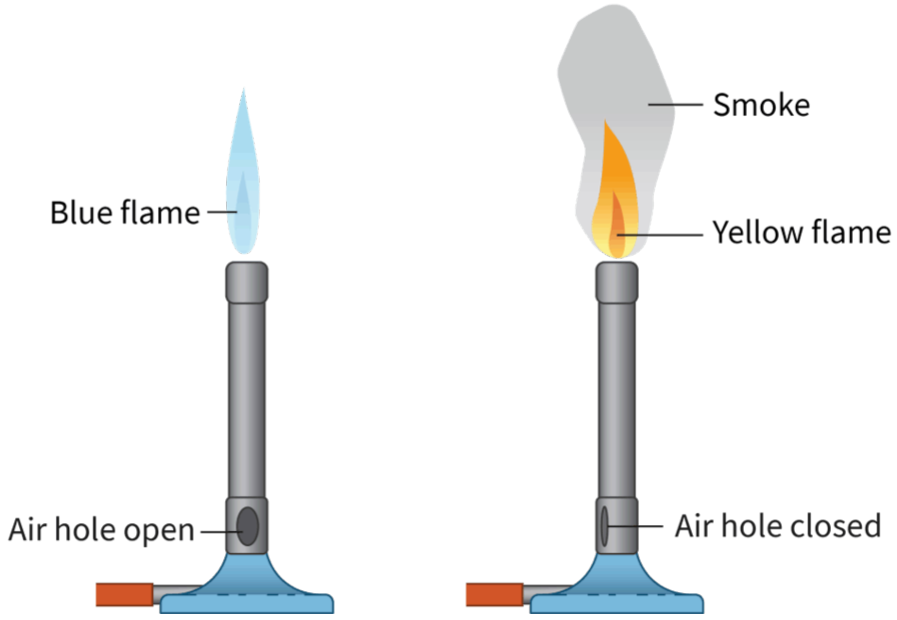

Figure 1. Complete vs. incomplete combustion of methane gas in a Bunsen burner.

More information for figure 1

The image illustrates two Bunsen burners side by side. On the left, the Bunsen burner has an open air hole, producing a blue flame. This indicates complete combustion, which requires sufficient oxygen, resulting in a hotter, smokeless flame. On the right, the Bunsen burner's air hole is closed, creating a yellow flame and emitting smoke, demonstrating incomplete combustion due to limited oxygen supply. Key features such as 'Blue flame,' 'Yellow flame,' 'Smoke,' 'Air hole open,' and 'Air hole closed' are labeled next to their respective elements in the illustration.

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Making connections

In sections S1.3.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/) and S.1.3.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-hydrogen-emission-spectrum-id-44042/) you discovered that the wavelength of light produced in an emission spectrum is inversely related to the energy of the wave. Yellow and orange wavelengths of visible light (≅550 nm) have relatively long wavelengths and therefore low energies. Blue light has a shorter wavelength (≅430 nm) and therefore a higher energy.

Examples of complete and incomplete combustion per mole of propane combusted

Sufficient oxygen supply:

$$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$$

Somewhat limited oxygen supply:

$$C_3H_8(g) + 3\tfrac{1}{2}O_2(g) \rightarrow 3CO(g) + 4H_2O(g)$$

Extremely limited oxygen supply:

$$C_3H_8(g) + 2O_2(g) \rightarrow 3C(s) + 4H_2O(g)$$

Nature of Science

Aspect: Global impact of science

The carbon particles produced during incomplete combustion reactions often appear as black or grey smoke and stay as a suspension in the air. Known as sooty by-products, carbon particles that are aerosolised into the atmosphere lead to a reduction in air quality and can have major impacts on respiratory health.

Carbon monoxide is an odourless, colourless, tasteless gas that is highly toxic and can lead to death if someone is exposed to even moderate concentrations for a prolonged period. Carbon monoxide molecules bind strongly to haemoglobin, in red blood cells, starving the body of oxygen.

The complete and incomplete combustion of carbon

Complete:

$$C(s) + O_2(g) \rightarrow CO_2(g) \quad \Delta H_{comb} \text{ is negative}$$

Incomplete:

$$2C(s) + O_2(g) \rightarrow 2CO(g) \quad \Delta H_{\mathrm{comb}} \text{ is negative}$$

Both reactions involve the same reactants and are exothermic, but in the incomplete combustion reaction, the oxygen is highly limited and only carbon monoxide is formed as a product instead of carbon dioxide. The product produced contains a lower ratio of oxygen to carbon atoms.

Incomplete combustion of hydrocarbons

All hydrocarbons undergo incomplete combustion reactions when reacted with limited amounts of oxygen gas. Take for example the incomplete combustion of propane, a common household and industrial fuel.

When the amount of oxygen is somewhat limited the reaction produces carbon monoxide gas and water vapour, as seen below:

$$C_3H_8(g) + 3\tfrac{1}{2}O_2(g) \rightarrow 3CO(g) + 4H_2O(g)$$

or balanced using whole-number coefficients:

$$2C_3H_8(g) + 7O_2(g) \rightarrow 6CO(g) + 8H_2O(g)$$

When the amount of oxygen is extremely limited the reaction produces carbon particles and water vapour:

$$C_3H_8(g) + 2O_2(g) \rightarrow 3C(s) + 4H_2O(g)$$

The same can be seen when looking at the incomplete combustion of alcohols. Take for example the incomplete combustion of ethanol:

$$2CH_3CH_2OH(l) + 4O_2(g) \rightarrow 4CO(g) + 6H_2O(g)$$

$$2CH_3CH_2OH(l) + 2O_2(g) \rightarrow 4C(s) + 6H_2O(g)$$

Study skills

You have learned that the quantity of oxygen will determine the products of an incomplete combustion reaction. Some incomplete combustion reactions produce a combination of carbon products in varying ratios, depending on the amount of oxygen present. These specific reaction equations will be balanced for you since the ratio of products can differ from one reaction to the next.

Worked example 1

The incomplete combustion of propanol takes place under very low amounts of oxygen. List the products for this type of incomplete combustion reaction.

Solid carbon particulates and water vapour. These are the two products produced from the incomplete combustion of an alcohol (like propanol) under very low oxygen concentrations.

Somewhat low oxygen concentrations would lead to incomplete combustion, where the products would be carbon monoxide and water vapour.

Sufficient oxygen concentrations would lead to complete combustion, which has carbon dioxide and water vapour as the products.

Activity

- **IB learner profile attribute:**
  - Inquirer
  - Reflective
- **Approaches to learning:**
  - Communication skills — Using terminology, symbols and communication conventions consistently and correctly.
  - Thinking skills — Using terminology, symbols and communication conventions consistently and correctly,
  - Research skills — Comparing, contrasting and validating information
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Pair activity

Instructions

In this activity you are asked to work with a partner to perform research on carbon monoxide detectors. You will need to use two or more reliable sources to support your answers to the following questions. The answers to these questions will help you to connect the topic of incomplete combustion to real life situations.

Recall that a reliable source is an article, journal, report, website etc. where you have confidence in the validity of the material. Examples include textbooks, reference books and databases, as well as government research documents and peer-reviewed reports.

1. List a variety of locations where it is necessary to install a carbon monoxide detector.
2. Describe the chemical(s) that these devices detect.
3. Discuss a particular type of carbon monoxide detector and how it functions. Identify the sensitivity of the chosen detector.
4. Explain why these types of detectors are essential in certain situations (types of heating) and not others. Support your answer by linking it to the incomplete combustion reaction for propane (a common heating fuel).